

Optical Amplification in the S-band and Beyond: From Materials and Mechanisms to Practical Deployment

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Abstract We review recent advances in S-band optical amplifiers for communication systems, comparing doped-fibre, Raman, parametric, and semiconductor approaches. Gain mechanisms, performance, and efficiency are analysed, highlighting key trade-offs and emerging solutions for scalable, energy-efficient transmission within and beyond the S-band. ©2026 The Author(s)

Introduction

The achievable capacity of commercial optical communication systems is approaching fundamental limits imposed by fibre nonlinearity and amplifier bandwidth, motivating the exploration of wider transmission spectra. Recent ultra-wideband (UWB) demonstrations spanning multiple spectral regions from the O- to the U-band have shown the potential for unprecedented capacity scaling [1–3]. However, such systems rely on complex combinations of amplification technologies and often require non-standard components, making their practical deployment challenging [4]. In contrast, conventional C- and L-band systems represent a mature and scalable solution, benefiting from highly optimised erbium-doped fibre amplifiers (EDFAs), well-established components, and extensive field deployment. Extending these systems toward wider bandwidths while maintaining similar levels of practicality remains a key challenge for next-generation networks.

The S-band, covering approximately 1460–1530 nm, offers a compelling pathway for incremental spectral expansion while leveraging the existing infrastructure. In this wavelength range, fibre attenuation remains low and compatible with standard silica-based infrastructure, enabling straightforward integration with existing systems. Unlike shorter-wavelength bands, the S-band does not impose significant constraints in terms of fibre loss or component availability, making it a promising candidate for near-term deployment. However, no single amplification technology in the S-band currently satisfies all performance and deployability requirements.

In this paper, we review recent advances in S-band optical amplifiers from both device- and system-level perspectives. First, we compare different gain media and amplifier architectures, analysing their performance, practical limitations, and key trade-offs relevant to deployment. In the

context of bandwidth expansion, S-band amplification is of significant interest. However, for certain applications—such as high-capacity multi-rail links—multi-band amplification schemes enabling seamless S- and C-band operation may represent a more practical and scalable technological step. Accordingly, we place particular emphasis on technologies that not only address the S-band in isolation, but also support co-propagation and integration with other bands, especially the C-band. Finally, we discuss emerging trends and their implications for extending optical communication systems beyond the S-band toward truly ultra-wideband operation.

Solutions

For a long time, the S-band has been considered the most practical next step for spectral expansion. Consequently, it has attracted a range of amplification approaches, with an early comprehensive review presented by Sakamoto at OFC 2001 [5]. At that time, four main solutions were identified at a comparable level of maturity: thulium-doped fibre amplifiers (TDFAs), gain-shifted TDFAs, and both lumped (LRA) and distributed Raman amplifiers (DRA).

Early TDFAs suffered from significant efficiency penalties due to the short lifetime of the upper laser level. This limitation was later mitigated by incorporating thulium ions into fluoride glass hosts [6]. In addition, the initially limited operational bandwidth was extended by employing two-stage TDFA configurations, enabling coverage of the full S-band. However, the use of fluoride fibres introduces practical challenges, including more difficult fibre handling and additional splice losses when interfacing with conventional silica-based components. These factors ultimately constrain the achievable noise figure (NF) and output power.

The remaining solutions—LRAs and DRAs—have followed different development paths. In recent years, DRAs have played a

Tab. 1: Comparison of S-band optical amplifier technologies

Type	Gain (dB)	NF (dB)	3-dB BW (nm)	PCE (%)	Input / loading	Remarks
TDFA [9]	14–19.5	5–7.5	~50	5–15	full load, -2.3 dBm	fluoride; gain-limited
DRA [9]	15–17.5	~3 eff	~50	5–10	1.4 dBm, distributed	low NF; complex
LRA [8]	10.5–16.8	5.8–7.6	~180	5–10	full load, -2.3 dBm	practical; moderate NF
S-band EDFA [10]	5.8–22.6	6–7	~40	10–20	-5 dBm, partial load	narrowband; multi-stage
SOA [11]	16.5–19.8	5.5–7.5	>100	30(e/o)	0 dBm, full load	compact; nonlinear
Dual-stage BDFA [12]	9–22.3	8.5–11.5	~140	10–25	5 dBm, single channel	silica; broadband
S-band BDFA [13]	11–21.5	6–7.5	~60	10–25	0 dBm, single channel	silica; broadband
Hybrid BDFA/EDFA [14]	15–33.5	5–7.2	~150	10–20	0 dBm, semi-full	no gain flatness; wideband
Hybrid BDFA/EDFA [15]	17–22	3.9–7	~75	10–20	0 dBm total, full load	flat gain; wideband
FOPA [16]	10.5–12.6	–	~15	< 5	~5 dBm, full load	phase-matched; narrow
Wideband OPA [17]	20–35	5–8	~60–100	< 5	~5 dBm, semi-loaded	high gain; complex

key role in multi-band transmission demonstrations, enabling extended reach and improved noise performance, particularly in higher-loss spectral regions [1, 7]. In contrast, LRAs can be implemented as discrete amplification units and have been explored as standalone solutions for multi-band systems [8]. However, while their NF can be comparable to that of TDFA, LRAs typically exhibit higher nonlinear noise and degraded performance relative to DRAs due to the use of long nonlinear fibres as the gain medium. A recent long-haul transmission comparison of LRA, DRA, and TDFA in the S-band demonstrated that DRA provides the best overall performance, followed by TDFA, with LRA exhibiting the lowest performance among the considered solutions [9].

A comparison of representative S-band amplification solutions is summarised in Table 1, where key performance metrics are evaluated for input powers of approximately 0 dBm. The table also distinguishes between different loading conditions, including fully loaded, single-channel, and partially loaded operation. A graphical comparison of gain and NF across these technologies is presented in Fig. 1.

An alternative approach was proposed by Ono *et al.* [10], utilising erbium-doped fibre pumped at 980 nm as a gain medium for S-band amplification. To suppress amplified spontaneous emission (ASE) in the C-band, multiple cascaded gain and filtering stages were employed. The demonstrated amplifier achieved gains exceeding 20 dB over the 1490–1520 nm range, with noise figures (NFs) of 6–7 dB. However, the complexity of the design, requiring nine stages, limits its practical applicability.

Semiconductor optical amplifiers (SOAs) have also been demonstrated as candidates for multi-band amplification, including operation within the S-band [11]. With bandwidths exceeding 100 nm, SOAs enabled some of the earliest wideband transmission demonstrations incorporating the S-band. Despite advantages such as compact size and high electro-optical (e/o) conversion efficiency, SOAs suffer from significant internal nonlinearities and effects such as patterning [18], which degrade signal quality.

Bismuth-doped fibre amplifiers (BDFAs) have

been proposed as a promising solution for amplification across multiple unconventional spectral bands, including the O-, E-, S-, and U-bands [4]. Unlike rare-earth-doped systems, bismuth exhibits the unique ability to form different bismuth active centres (BACs), enabling gain in distinct spectral regions depending on the host glass composition [19].

One of the first demonstrations of a germanosilicate BDFA showed the potential for E- and S-band amplification, although with limited gain within S-band in the 1460–1490 nm range [20]. Subsequent studies investigated the impact of fibre composition and the use of dual-stage amplifier configurations to shift the gain further into the S-band [12]. The resulting amplifier achieved a broad bandwidth of approximately 100 nm, with gains ranging from 10 to 20 dB across 1460–1520 nm, particularly enhancing performance on the shorter-wavelength side of the S-band. These results were obtained under single-channel input conditions of 5 dBm, which limited the achievable noise figure (NF) to 8–10 dB due to moderate gain and nonlinear effects, such as pump–signal four-wave mixing, as discussed in detail in [21].

To further shift the gain bandwidth deeper into the S-band, a BDFA employing an ultra-long active fibre (~400 m) and long-wavelength pumping at 1350 nm and 1425 nm was proposed [13]. Under single-channel operation with an input power of 0 dBm, the amplifier achieved gains of 10–20 dB across the full S-band, with noise figures of 5–7 dB, comparable to state-of-the-art S-band TDFA.

Hybrid amplification approaches combining BDFAs and EDFAs have been proposed to enable seamless operation across the S- and C-bands while maintaining practical system performance. An initial demonstration by Maes *et al.* [14] introduced a multi-stage BDFA/EDFA architecture, where multiple cascaded gain blocks were used to achieve wideband amplification spanning the E-, S-, and C-bands. These designs demonstrated gains exceeding 25 dB over bandwidths approaching 80–90 nm. However, the proposed configurations relied on relatively complex multi-stage implementations, typically involving four gain stages without providing flat-top gain. A subsequent study

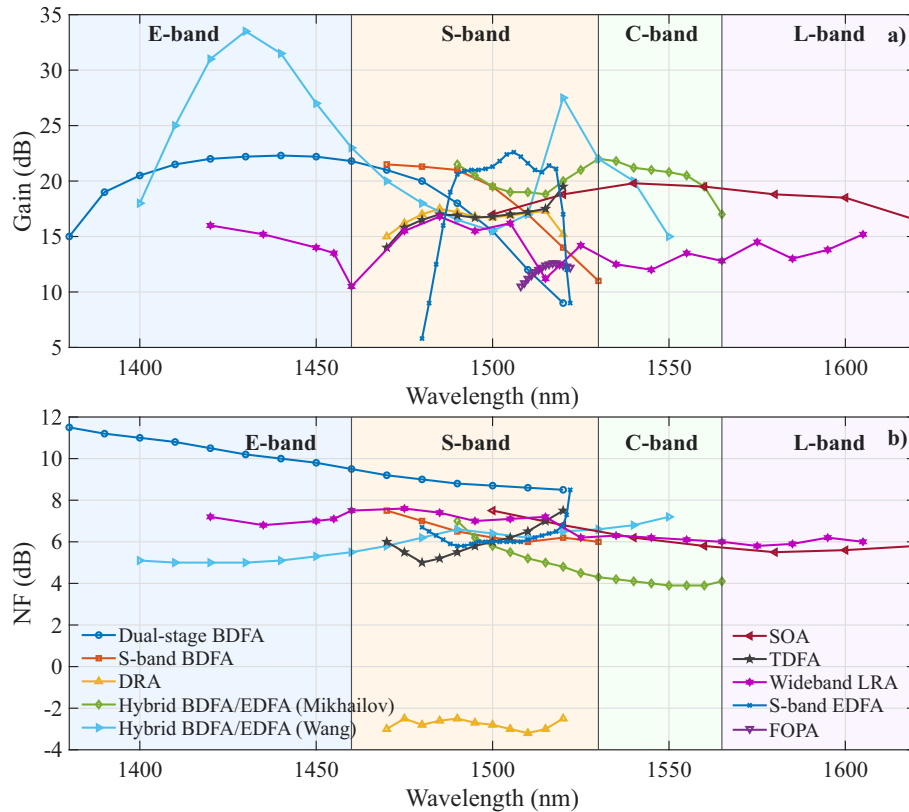


Fig. 1: (a) Gain and (b) noise figure spectra of amplifier configurations exhibiting S-band gain, shown across the E-, S-, C-, and L-bands.

by Wang *et al.* [22], proposed a similar design with reduced fibre length while testing the amplifiers under semi-loaded conditions.

The recent work by Mikhailov *et al.* [15] simplified this approach by realising a two-stage hybrid BDFA/EDFA amplifier incorporating a gain-flattening filter between the stages. This architecture enabled fully-loaded operation with a flat-top gain profile across the S- and C-bands, while significantly reducing system complexity compared to earlier multi-stage designs. In addition, improved noise performance was achieved, with noise figures reaching 4–5 dB, representing the lowest reported NF among the lumped S-band amplification solutions reported to date. As summarised in Table 1, the hybrid BDFA/EDFA approach provides a compelling balance between bandwidth, gain, and practical deployability, particularly for systems requiring seamless multi-band operation.

Parametric amplification represents an alternative approach based on optical Kerr nonlinearity. Fibre optical parametric amplifiers (FOPAs) operate through four-wave mixing (FWM), where a strong pump transfers energy to signal and idler waves under phase-matching conditions. In the S-band, FOPAs have demonstrated gains in the range of 10–18 dB with bandwidths of approximately 15–25 nm [16]. However, their performance is highly sensitive to phase-matching conditions, limiting their operational bandwidth and requiring careful dispersion engineering. In addition, rela-

tively low power conversion efficiency (PCE) and the need for high pump powers restrict their practical deployment.

Optical parametric amplifiers (OPAs), particularly those implemented in integrated photonic platforms, have recently demonstrated significantly broader bandwidths and higher gains compared to fibre-based FOPAs. Wideband OPAs with gains exceeding 20–30 dB over bandwidths of 60–100 nm have been reported [17]. Despite these promising results, experimental demonstrations in the S-band remain limited, and to date, performance has primarily been characterised in terms of conversion efficiency rather than full system-level amplification metrics. As a result, while parametric amplification offers significant potential for ultra-wideband operation, its applicability to practical S-band communication systems remains an open challenge.

Conclusion

The S-band offers a practical pathway for bandwidth expansion using existing infrastructure, but no single amplification technology fully meets all performance and deployability requirements. Among the explored solutions, hybrid BDFA/EDFA amplifiers provide the most compelling balance of bandwidth, gain, noise performance, and simplicity. Continued development of such practical amplifier architectures will be key to enabling scalable multi-band optical communication systems.

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